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UNIT 3
PLANT PHYSIOLOGY

Plant physiology is the branch of biology that deals with the internal activities of plants. It covers many aspects of plant life, including nutrition (photosynthesis), movement of materials around the plant, and reproduction. There are many reasons why we need to study how plants work, not least because they provide the world's food supply and oxygen. Animals, including humans, could not exist without plants. Modern agriculture depends on the scientific study of plant physiology, as does our understanding of eccosystems.



10 PLANTS AND FOOD

This chapter looks at photosynthesis, the process by which plants make starch, and the structure of leaves in relation to photosynthesis. It also deals with how plants obtain other nutrients, and their uses in the plant.

LEARNING OBJECTIVES

- Investigate photosynthesis by testing for the production of starch in a leaf
- Investigate the need for light, carbon dioxide and chlorophyll for photosynthesis
- Understand the process of photosynthesis and its importance in the conversion of light energy to chemical energy
- Know the word equation and the balanced symbol equation for photosynthesis
- Describe the structure of the leaf and explain how it is adapted for photosynthesis

BIOLOGY ONLY

- Understand how the structure of the leaf is adapted for gas exchange
- Understand the role of diffusion and stomata in gas exchange by a leaf

BIOLOGY ONLY

- Understand how the exchange of carbon dioxide and oxygen in plants depends on the balance between respiration and photosynthesis
- Understand how respiration in plants continues during the day and night, and that the net exchange of carbon dioxide and oxygen depends on the intensity of light
- Understand how light intensity, carbon dioxide concentration and temperature affect the rate of photosynthesis

BIOLOGY ONLY

- Investigate the effect of light on net gas exchange from a leaf, using hydrogen carbonate indicator
- Investigate photosynthesis using the evolution of oxygen from a water plant
- Understand that plants require mineral ions for growth, and that magnesium ions are needed to make chlorophyll and nitrate ions are needed to make amino acids

PLANTS MAKE STARCH

All the foods shown in Figure 10.1 are products of plants. Some, such as potatoes, rice and bread (made from cereals such as wheat or rye), form the staple diet of humans. They all contain starch, which is the main storage carbohydrate made by plants. Starch is a good way of storing carbohydrate because it insoluble, compact and can be broken down easily.



Figure 10.1 All these foods are made by plants or made from products of plants. They all contain starch.

ACTIVITY 1

▼ PRACTICAL: TESTING LEAVES FOR STARCH

You can test for starch in food by adding a few drops of yellow-brown iodine solution (see Chapter 4). If the food contains starch, a blue-black colour is produced.

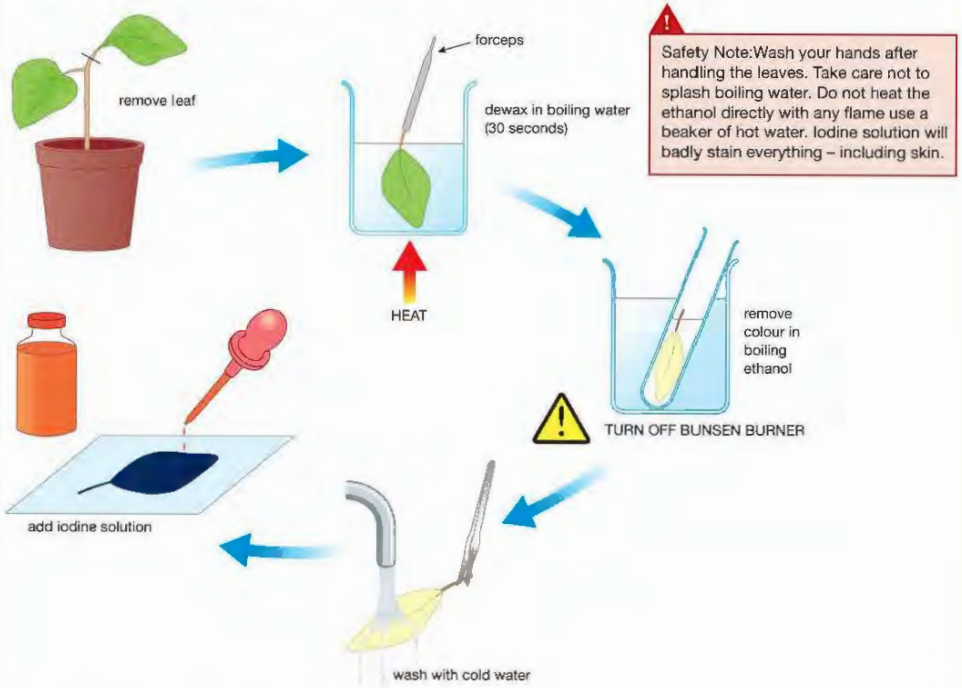
Leaves that have been in sunlight also contain starch, but you can't test for it by adding iodine solution to a fresh leaf. The outer waxy surface of the leaf will not absorb the solution, and besides, the green colour of the leaf would hide the colour change. To test for starch in a leaf, the outer waxy layer needs to be removed and the leaf decolourised. This is done by placing the leaf in boiling ethanol (see Figure 10.2).

A beaker of water is set up on a tripod and gauze and the water heated until it boils. A leaf is removed from a plant and killed by placing it in boiling water for 30 seconds (this stops all chemical reactions in the leaf).

The Bunsen burner is turned off (this is important because ethanol is highly flammable), the leaf is placed in a boiling tube containing ethanol, and the tube is placed in the beaker of hot water. The boiling point of ethanol (about 78 °C) is lower than that of water (100 °C) so the ethanol will boil for a few minutes, until the water cools down. This is long enough to remove most of the chlorophyll from the leaf.

When the leaf has turned colourless or pale yellow, it is removed and washed in cold water to soften it, then spread out on a tile and covered with a few drops of iodine solution. After a few minutes, any parts of the leaf that contain starch will turn a dark blue-black colour. This only works if the plant has had plenty of light for some hours before the test.

Safety Note: Wash your hands after handling the leaves. Take care not to splash boiling water. Do not heat the ethanol directly with any flame use a beaker of hot water. Iodine solution will badly stain everything – including skin.

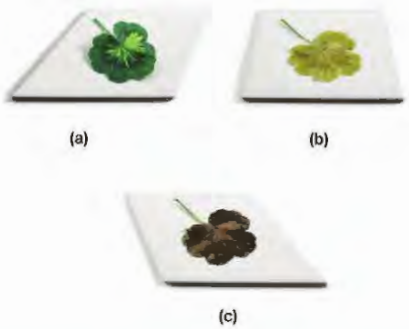


▲ Figure 10.2 How to test a leaf for starch

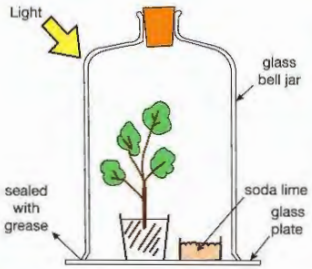
KEY POINT

You can 'de-starch' a plant by placing it in the dark for 2 or 3 days. The plant uses up the starch stores in its leaves. De-starched plants are used to find out the conditions needed for the plant to make more starch by photosynthesis.

Starch is only made in the parts of plants that contain chlorophyll. You can show this by testing a leaf from a variegated plant, which has green and white areas to its leaves. The white parts of the leaf will give a negative starch test, staining yellow-brown with iodine solution. Only the green areas will stain blue-black. Figure 10.3 shows the results of a starch test on a leaf. The leaf was taken from a plant that had been under a bright light for 24 hours.



▲ Figure 10.3 Testing a leaf for starch. (a) Leaf before test (b) Decolourised leaf (c) Leaf after test, stained blue-black with iodine solution.



▲ Figure 10.4 Demonstration that carbon dioxide is needed for photosynthesis. The soda lime absorbs carbon dioxide from the air in the bell jar. A Control experiment should be set up, using exactly the same apparatus but without the soda lime.

HINT

You might think that the results of the test on the variegated leaf prove that chlorophyll is needed for photosynthesis. But is this conclusion fully justified? The leaf could have photosynthesised in the white areas and the sugars been transported elsewhere in the plant. Similarly, the green areas may not be photosynthesising at all, but simply laying down starch from glucose made somewhere else. All it really shows is that starch is present in the green areas and not in the white areas of the leaf. We *assume* this is because chlorophyll is needed for photosynthesis.

Taking away the source of light is not the only way you can prevent a plant making starch in its leaves. You can also place it in a closed container containing a chemical called soda lime (Figure 10.4). This substance absorbs carbon dioxide from the air around the plant. If the plant is kept under a bright light but with no carbon dioxide, it again won't be able to make starch.

WHERE DOES THE STARCH COME FROM?

You have now found out three important facts about starch production by leaves:

- it uses carbon dioxide from the air
- it needs light
- it needs chlorophyll in the leaves.

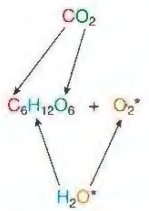
As well as starch, there is another product of this process which is essential to the existence of most living things on the Earth – oxygen. When a plant is in the light, it makes oxygen gas. You can show this using an aquatic plant such as *Elodea* (Canadian pondweed). When a piece of this plant is placed in a test tube of water under a bright light, it produces a stream of small bubbles. If the bubbles are collected and their contents analysed, they are found to contain a high concentration of oxygen (Figure 10.5).



▲ Figure 10.5 The bubbles of gas released from this pondweed contain a higher concentration of oxygen than in atmospheric air.

Starch is composed of long chains of glucose (see Chapter 4). A plant does not make starch directly, but first produces glucose, which is then joined together in chains to form starch molecules. A carbohydrate made of many sugar sub-units is called a **polysaccharide**. Glucose has the formula $C_6H_{12}O_6$. The carbon and oxygen atoms of the glucose molecule come from the carbon dioxide gas in the air around the plant. The hydrogen atoms come from another molecule essential to the living plant – water.

It would be very difficult in a school laboratory to show that a plant uses water to make starch. If you deprived a plant of water in the same way as you deprived it of carbon dioxide, it would soon wilt and eventually die. However, scientists have proved that water is used in photosynthesis. They have done this by supplying the plant with water with 'labelled' atoms, for example using the 'heavy' isotope of oxygen (^{18}O). This isotope ends up in the oxygen gas produced by the plant. A summary of the sources of the atoms in the glucose and oxygen looks like this:



(*oxygen labelled with ^{18}O)

EXTENSION WORK

Isotopes are forms of the same element with the same atomic number but different mass numbers (due to extra neutrons in the nucleus). Isotopes of some elements are radioactive and can be used as 'labels' to follow chemical pathways. Others like ^{18}O are identified by their mass.

PHOTOSYNTHESIS

Plants use the simple inorganic molecules carbon dioxide and water, in the presence of chlorophyll and light, to make glucose and oxygen. This process is called **photosynthesis**.

It is summarised by the equation:

carbon dioxide + water $\xrightarrow{\text{light}}$ glucose + oxygen

or: $6CO_2 + 6H_2O \xrightarrow{\text{chlorophyll}} C_6H_{12}O_6 + 6O_2$

DID YOU KNOW?

The 'photo' in photosynthesis comes from the Greek word *photos*, meaning light, and a 'synthesis' reaction is one where small molecules are built up into larger ones.

The role of the green pigment, chlorophyll, is to absorb the light energy needed for the reaction to take place. The products of the reaction (glucose and oxygen) contain more energy than the carbon dioxide and water. In other words, photosynthesis converts light energy into chemical energy.

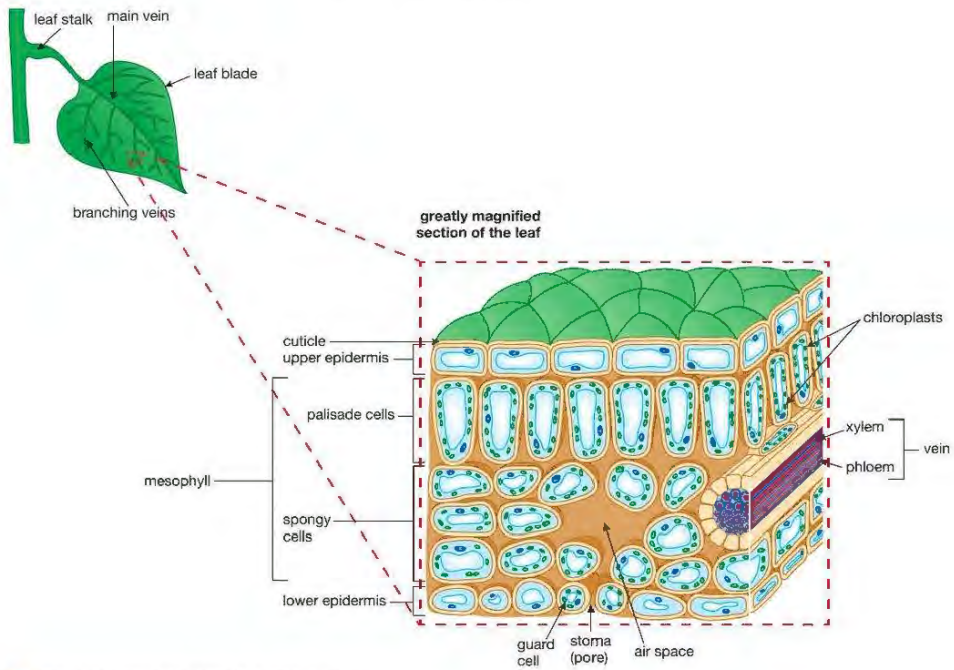
You will probably have noticed that the overall equation for photosynthesis is the reverse of the one for aerobic respiration (see Chapter 1):

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (plus energy)

Respiration, which is carried out by both animals and plants, *releases* energy (but not as light) from the breakdown of glucose. The chemical energy in the glucose came originally from light 'trapped' by the process of photosynthesis.

THE STRUCTURE OF LEAVES

Most green parts of a plant can photosynthesise, but the leaves are the plant organs which are best adapted for this function. To be able to photosynthesise efficiently, leaves need to have a large surface area to absorb light, many chloroplasts containing the chlorophyll, a supply of water and carbon dioxide, and a system for carrying away the products of photosynthesis to other parts of the plant. They also need to release oxygen (and water vapour) from the leaf cells. Most leaves are thin, flat structures supported by a leaf stalk which can grow to allow the blade of the leaf to be angled to receive the maximum amount of sunlight (Figure 10.6).



▲ Figure 10.6 External and internal features of a leaf

KEY POINT

You will only need to know about adaptations for a gas exchange in a leaf if you are taking the Biology course; but not if you are taking the Double Award course.

Inside the leaf are layers of cells with different functions.

- The two outer layers of cells (the upper and lower **epidermis**) have few chloroplasts and are covered by a thin layer of a waxy material called the **cuticle**. This reduces water loss by evaporation, and acts as a barrier to the entry of disease-causing microorganisms such as bacteria and fungi.
- The lower epidermis has many holes or pores called **stomata** (a single pore is a **stoma**). Usually the upper epidermis contains fewer or no stomata. The stomata allow carbon dioxide to diffuse into the leaf, to reach the photosynthetic tissues. They also allow oxygen and water vapour to diffuse out. Each stoma is formed as a gap between two highly specialised cells called **guard cells**, which can change their shape to open or close the stoma (see Chapter 11).

KEY POINT

A 'gas exchange' surface is a tissue that allows gases (usually oxygen, carbon dioxide and water vapour) to pass across it between the plant or animal and the outer environment. Gas exchange surfaces all have a large surface area in proportion to their volume, which allows large amounts of gases to diffuse across. Examples include the alveoli of the lungs, the gills of a fish and the spongy mesophyll of a leaf.

KEY POINT

Starch is insoluble and so cannot be transported around the plant. The phloem carries only soluble substances such as sugars (mainly sucrose) and amino acids. These are converted into other compounds when they reach their destination.

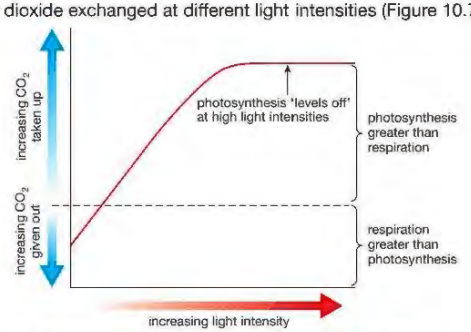
- In the middle of the leaf are two layers of photosynthetic cells called the mesophyll ('mesophyll' just means 'middle of the leaf'). Just below the upper epidermis is the **palisade mesophyll** layer. This is a tissue made of long, narrow cells, each containing hundreds of chloroplasts, and is the main site of photosynthesis. The palisade cells are close to the source of light, and the upper epidermis is relatively transparent, allowing light to pass through to the enormous numbers of chloroplasts which lie below.
- Below the palisade layer is a tissue made of more rounded, loosely packed cells, with air spaces between them, called the **spongy mesophyll** layer. These cells also photosynthesise, but have fewer chloroplasts than the palisade cells. They form the main gas exchange surface of the leaf, absorbing carbon dioxide and releasing oxygen and water vapour. The air spaces allow these gases to diffuse in and out of the mesophyll.
- Water and mineral ions are supplied to the leaf by vessels in a tissue called the **xylem**. This forms a continuous transport system throughout the plant. Water is absorbed by the roots and passes up through the stem and through veins in the leaves in the **transpiration stream**. In the leaves, the water leaves the xylem and supplies the mesophyll cells.
- The products of photosynthesis, such as sugars, are carried away from the mesophyll cells by another transport system, the **phloem**. The phloem supplies all other parts of the plant, so that tissues and organs that can't make their own food receive products of photosynthesis. The veins in the leaf contain both xylem and phloem tissue, and branch again and again to supply all parts of the leaf.

You will find out more about both plant transport systems in Chapter 11.

BIOLOGY ONLY

PHOTOSYNTHESIS AND RESPIRATION

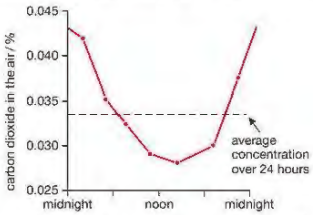
Through photosynthesis, plants supply animals with two of their essential needs – food and oxygen – as well as removing carbon dioxide from the air. But remember that living cells, including plant cells, respire *all the time*, and they need oxygen for this. When the light intensity is high, a plant carries out photosynthesis at a much higher rate than it respires. So in bright light, there is an overall uptake of carbon dioxide from the air around a plant's leaves, and a surplus production of oxygen that animals can use. A plant only produces more carbon dioxide than it uses up in dim light. We can show this as a graph of carbon dioxide exchanged at different light intensities (Figure 10.7).



The point where the curve crosses the dashed line shows where photosynthesis is equal to respiration – there is no net gain or loss of CO_2.

▲ Figure 10.7 As the light intensity gets higher, photosynthesis speeds up, but eventually levels off in very bright light.

The concentration of carbon dioxide in the air around plants actually changes throughout the day. Scientists have measured the level of carbon dioxide in the air in the middle of a field of long grass in summer. They found that the air contained least carbon dioxide in the afternoon, when photosynthesis was happening at its highest rate (Figure 10.8). At night, when there was no photosynthesis, the level of carbon dioxide rose. This rise is due to less carbon dioxide being absorbed by the plants, while carbon dioxide was added to the air from the respiration of all organisms in the field.



▲ Figure 10.8 Photosynthesis affects the concentration of carbon dioxide in the air around plants. Over a 24-hour period, the concentration rises and falls, as a result of the relative levels of photosynthesis and respiration.

INVESTIGATING THE EFFECT OF LIGHT ON GAS EXCHANGE BY A LEAF

KEY POINT

'Equilibration' means allowing something to reach a 'balanced state'. In this example the solution is left exposed to atmospheric air until the movement of CO_2 into the solution is equal to the movement of CO_2 out. (We also talk about 'equilibrating a liquid to temperature', which means leaving the liquid until it reaches the surrounding temperature – when the movement of heat in is equal to the movement of heat out).

Hydrogen carbonate indicator solution is very sensitive to changes in carbon dioxide concentration. When the solution is made, it is equilibrated with atmospheric air, which has a CO_2 concentration of 0.04%. If extra CO_2 is added to the solution, or if CO_2 is taken away from the solution, it changes colour (Table 10.1).

Table 10.1 Changes to hydrogen carbonate indicator solution with different concentrations of carbon dioxide.

Condition	Indicator colour
high concentrations of CO_2 (more than 0.04%)	yellow
CO_2 in normal air (0.04%)	orange
low concentrations of CO_2 (less than 0.04%)	purple

! Safety Note: Wash hands after collecting and inserting leaves.

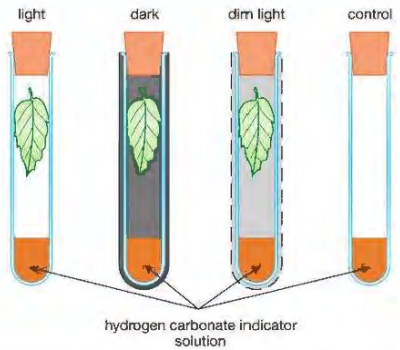
ACTIVITY 2

▼ PRACTICAL: INVESTIGATING THE EFFECT OF LIGHT ON GAS EXCHANGE BY A LEAF

Hydrogen carbonate indicator solution can be used to show the effect of light on the exchange of carbon dioxide by a leaf.

A volume of 10 cm^3 of the indicator solution is placed in the bottom of each of four boiling tubes. Three large leaves are detached from a suitable plant and placed in the tops of three of the tubes. Each tube is sealed with a rubber bung.

One tube is placed in bright light, one in the dark (covered with silver foil) and the third covered with some translucent material, to restrict the intensity of light reaching the leaf. The fourth tube without a leaf acts as a Control, to show that it is the presence of the leaf which affects the colour of the indicator. (Figure 10.9).



▲ Figure 10.9 Using hydrogen carbonate indicator to test the effect of light on gas exchange.

The four tubes are left set up under a bright light for a few hours, and the colour of the indicator recorded. The results from one experiment like this are shown below.

Tube	Colour of indicator at end of experiment
leaf in light	purple
leaf in dark	Yellow
leaf in dim light	orange
control (no leaf)	orange

Use your knowledge of a plant's rate of respiration and photosynthesis in the light and dark to explain these results. You may be able to carry out a similar investigation.

END OF BIOLOGY ONLY

FACTORS AFFECTING THE RATE OF PHOTOSYNTHESIS

KEY POINT

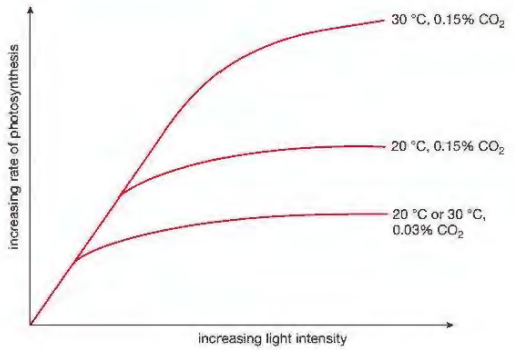
A limiting factor is the component of a reaction that is in 'shortest supply' so that it prevents the rate of the reaction increasing, in other words sets a 'limit' to it.

In Figure 10.7, you can see that when the light intensity rises, the rate of photosynthesis begins by rising too, but eventually it reaches a maximum rate. What makes the rate 'level off' (flatten) like this? It is because some other factor needed for photosynthesis is in short supply, so that increasing the light intensity does not affect the rate any more. Normally, the factor which 'holds back' the rate of photosynthesis is the concentration of carbon dioxide in the air. This is only about 0.03 to 0.04%, and the plant can only take up the carbon dioxide and fix it into carbohydrate at a certain rate. If the plant is put in a closed container with a higher than normal concentration of carbon dioxide, it will photosynthesise at a faster rate. If there is both a high light intensity and a high level of carbon dioxide, the temperature may limit the rate of photosynthesis, by limiting the rate of the chemical reactions in the leaf. A rise in temperature will then increase the rate. With normal levels of carbon dioxide, very low temperatures (close to 0 °C) slow the reactions, but high temperatures (above about 35 °C) also reduce photosynthesis by denaturing enzymes in the plant cells (see Chapter 1).

KEY POINT

Knowledge of limiting factors is used in some glasshouses (greenhouses) to speed up the growth of crop plants such as tomatoes and lettuces (see Chapter 15). Extra carbon dioxide is added to the air around the plants, by using gas burners. The higher concentration of carbon dioxide, along with the high temperature in the glasshouse, increases the rate of photosynthesis and the growth of the leaves and fruits.

Light intensity, carbon dioxide concentration and temperature can all act as what are called **limiting factors** in this way. This is easier to see as a graph (Figure 10.10).



▲ Figure 10.10 Light intensity, carbon dioxide concentration and temperature can all act as limiting factors on the rate of photosynthesis.

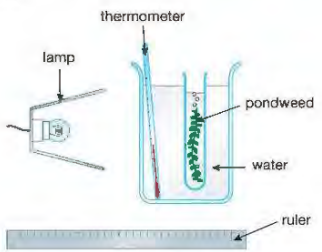
! Safety Note: Wash hands after collecting and preparing pondweed. Do not handle the lamp, plug or switch with wet hands.

ACTIVITY 3

▼ PRACTICAL: MEASURING THE RATE OF PHOTOSYNTHESIS USING PONDWEED

You can measure the rate of photosynthesis of a plant by measuring how quickly it produces oxygen. With a land plant this is difficult, because the oxygen is released into the air, but with an aquatic plant such as the pondweed *Elodea*, bubbles of oxygen are released into the water around the plant (Figure 10.5).

If the bubbles formed per minute are counted, this is a measure of the rate of photosynthesis of the plant. It is easiest to count the bubbles if the cut piece of weed is placed upside down in a test tube, as shown in Figure 10.11. A small paperclip attached to the bottom of the piece of weed makes it sink.



▲ Figure 10.11 Measuring the rate of photosynthesis in an aquatic plant.

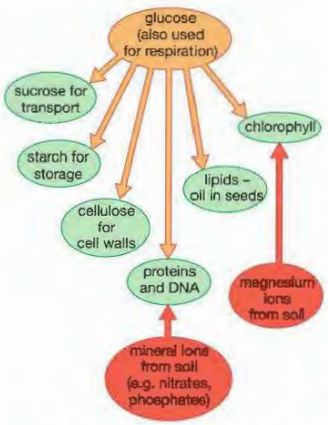
The light intensity is changed by moving the lamp, altering the distance between the lamp and the pondweed. The beaker of water keeps the temperature of the plant constant.

Design an experiment using this apparatus to find out if the rate of photosynthesis is affected by the light intensity. In your plan you should include:

- a hypothesis – state what you think will happen when you change the light intensity, and why
- a systematic way of changing the light intensity
- how the experiment will be controlled so that nothing else is changed apart from the light intensity (e.g. what will you do about the background light in the laboratory?)
- a Control that you could use, to show that it is the effect of light on the pondweed that is producing the bubbles
- how you will ensure that your results are reliable.

When you have completed your plan, you may be allowed to use similar apparatus to carry out the experiment. How could you modify your plan to find the effect of changing the *temperature* on the rate of photosynthesis? What factors would you need to keep constant this time? What would be a suitable range of temperatures to use?

THE PLANT'S USES FOR GLUCOSE



▲ Figure 10.12 Compounds that plant cells can make from glucose.

As you have seen, some glucose that the plant makes is used in respiration to provide the plant's cells with energy. Some glucose is quickly converted into starch for storage. However, a plant is not made up of just glucose and starch, and must make all of its organic molecules, starting from glucose.

Glucose is a single sugar unit (a **monosaccharide**). Plant cells can convert it into other sugars, such as a monosaccharide called **fructose** (found in fruits) and the **disaccharide sucrose**, which is the main sugar carried in the phloem. It can also be changed into another polymer, the polysaccharide called **cellulose**, which forms plant cell walls.

All these compounds are carbohydrates. Plant cells can also convert glucose into lipids (fats and oils). Lipids are needed for the membranes of all cells, and are also an energy store in many seeds and fruits, such as peanuts, sunflower seeds and olives.

Carbohydrates and lipids both contain only three elements – carbon, hydrogen and oxygen – and so they can be inter-converted without the need for a supply of other elements. Proteins contain these elements too, but all amino acids (the building blocks of proteins) also contain nitrogen. This is obtained as nitrate ions from the soil, through the plant's roots. Other compounds in plants contain other elements. For example, chlorophyll contains magnesium ions, which are also absorbed from water in the soil. Some of the products that a plant makes from glucose are summarised in Figure 10.12.

KEY POINT

Glucose from photosynthesis is not just used as the *raw material* for the production of molecules such as starch, cellulose, lipids and proteins. Reactions like these, which synthesise large molecules from smaller ones, also need a source of energy. This energy is provided by the plant's *respiration* of glucose.

MINERAL NUTRITION

Nitrate ions are absorbed from the soil water, along with other minerals such as phosphate, potassium and magnesium ions. The element phosphorus is needed for the plant cells to make many important compounds, including DNA. Potassium ions are required for enzymes in respiration and photosynthesis to work, and magnesium forms a part of the chlorophyll molecule.

WATER CULTURE EXPERIMENTS

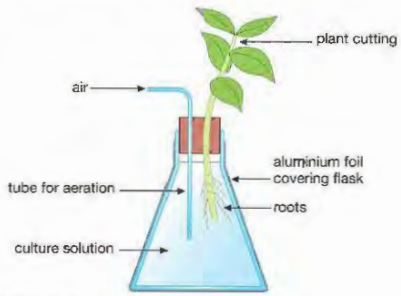
A plant takes only water and mineral ions from the soil for growth. Plants can be grown in soil-free cultures (water cultures) if the correct balance of minerals is added to the water. In the nineteenth century, the German biologist Wilhelm Knop invented one example of a culture solution. Knop's solution contains the following chemicals (per dm^3 of water):

0.8 g	calcium nitrate
0.2 g	magnesium sulfate
0.2 g	potassium nitrate
0.2 g	potassium dihydrogenphosphate
(trace)	iron(III) phosphate

DID YOU KNOW?

In fact, in addition to the ions listed in Knop's solution, plants need very small amounts of other mineral ions for healthy growth. Knop's culture solution only worked because the chemicals he used to make his solutions weren't very pure, and supplied enough of these additional ions by mistake!

Notice that these chemicals provide all of the main elements that the plant needs to make proteins, DNA and chlorophyll, as well as other compounds, from glucose. It is called a *complete* culture solution. If you were to make a similar solution, but to replace, for example, magnesium sulfate with more calcium sulfate, this would produce a culture solution which was *deficient* (lacking) in magnesium. You could then grow plants in the complete and deficient solutions, and compare the results. There are several ways to grow the plants, such as using the apparatus shown in Figure 10.13, which is useful for plant cuttings. Seedlings can be grown by packing cotton wool around the seed, instead of using a rubber bung.



▲ Figure 10.13 A simple water culture method

The plant is kept in bright light, so that it can photosynthesise. The covering around the flask prevents algae from growing in the culture solution. The aeration tube is used for short periods to supply the roots with oxygen for respiration of the root cells, since some ions are taken up by active transport. Using methods like this, it soon becomes clear that mineral deficiencies result in poor plant growth. A shortage of a particular mineral results in particular



▲ Figure 10.14 (a) A healthy bean plant

symptoms in the plant, called a *mineral deficiency disease*. For example, lack of magnesium means that the plant won't be able to make chlorophyll, and the leaves will turn yellow. Lack of nitrate limits a plant's growth, because it is unable to make enough protein. Some of the mineral ions that a plant needs, their uses, and the deficiency symptoms are shown in Table 10.2. Compare the photographs of the mineral deficient plants to those of the control plant in Figure 10.14(a).

Table 10.2: Mineral ions needed by plants

Mineral ion	Use	Deficiency symptoms
nitrate	making amino acids, proteins, chlorophyll, DNA and many other compounds	limited growth of plant; older leaves turn yellow  ▲ Figure 10.14 (b) A plant showing symptoms of nitrate deficiency.
phosphate	making DNA and many other compounds; part of cell membranes	poor root growth; younger leaves turn purple  ▲ Figure 10.14 (c) A plant showing symptoms of phosphate deficiency.
potassium	needed for enzymes of respiration and photosynthesis to work	leaves turn yellow with dead spots  ▲ Figure 10.14 (d) A plant showing symptoms of potassium deficiency.
magnesium	part of chlorophyll molecule	leaves turn yellow  ▲ Figure 10.14 (e) A plant showing symptoms of magnesium deficiency.

DID YOU KNOW?

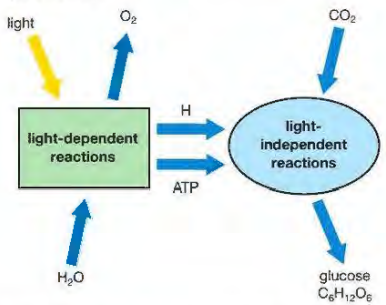
Some commercial crops such as lettuces can be grown without soil in culture solutions. This is called **hydroponics**. The plants' roots grow in a long plastic tube that has culture solution passing through it (Figure 10.15). The composition of the solution can be carefully adjusted to ensure the plants grow well. Pests such as insects, which might live in soil, are also less of a problem.



▲ Figure 10.15 Lettuce plants grown by hydroponics

LOOKING AHEAD – MORE ABOUT PHOTOSYNTHESIS

The role of chlorophyll is to absorb the light energy needed to drive photosynthesis. The products of the reaction (glucose and oxygen) contain more chemical energy than the reactants (carbon dioxide and water). The simple equation for photosynthesis that you saw earlier in this chapter is actually a summary of a series of reactions. They take place in two stages (Figure 10.16).



▲ Figure 10.16 Photosynthesis takes place in two stages called light-dependent and light-independent reactions.

The light energy absorbed by the chlorophyll is used to split water molecules into hydrogen and oxygen. The waste product, oxygen, is given off as a gas. At the same time, the light energy is used to convert ADP and phosphate into ATP. Because this stage needs light, the steps in the process are known as the *light-dependent reactions*.

Next, the hydrogen atoms from the water and energy from the ATP are used to reduce carbon dioxide to glucose. This takes place in another series of reactions in the chloroplast. Because these do not need light, they are called the *light-independent reactions*.

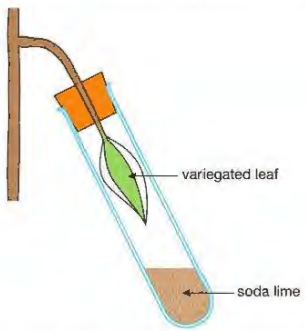
If you continue with biology beyond International GCSE, you will find out more about the biochemistry of photosynthesis.

CHAPTER QUESTIONS

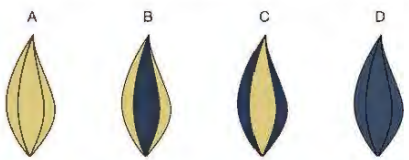
More questions on plants and food can be found at the end of Unit 3 on page 182.

SKILLS REASONING (7)

1 A variegated leaf attached to a plant was placed in a test tube as shown below. The test tube contained soda lime to absorb carbon dioxide.



The plant was left in bright light for 24 hours and tested for starch using iodine solution. Which of the following diagrams shows the expected results?



BIOLOGY ONLY

SKILLS CRITICAL THINKING (6)

2 At night, the concentration of carbon dioxide in the air in a forest is high. What is the main reason for this?

A There is increased respiration by animals that are active at night

B Plants are unable to absorb carbon dioxide because their stomata are shut

C Plants begin to respire in the dark

D Plants can only use carbon dioxide for photosynthesis in the light

END OF BIOLOGY ONLY

(5)

3 Which of the following is not normally a factor that limits the rate of photosynthesis?

A temperature

B oxygen concentration

C carbon dioxide concentration

D light intensity

SKILLS CRITICAL THINKING (5)

4 Which of the following compounds can only be made by a plant if it has access to magnesium ions?

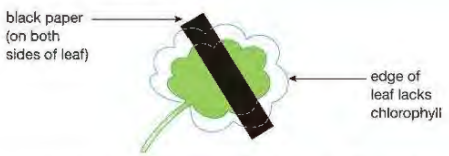
A chlorophyll

B cellulose

C starch

D protein

5 A plant with variegated leaves had a piece of black paper attached to one leaf as shown in the diagram.



The plant was kept under a bright light for 24 hours. The leaf was then removed, the paper taken off and the leaf was tested for starch.

a Name the chemical used to test for starch, and describe the colour change if the test is positive.

(6)

b Copy the leaf outline and shade in the areas which would contain starch.

c Explain how you arrived at your answer to (b).

SKILLS REASONING (5)

d What is starch used for in a plant? How do the properties of starch make it suitable for this function?

SKILLS INTERPRETATION (5)

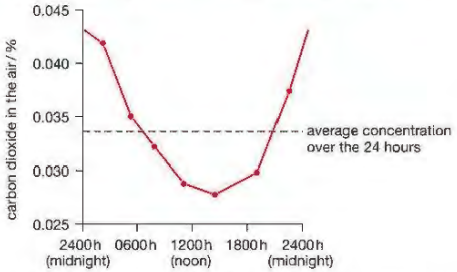
6 Copy and complete the following table to show the functions of different parts of a leaf, and how that part is adapted for its function. One row has been done for you.

Part of leaf	Function	How the part is adapted for its function
palisade mesophyll layer	main site of photosynthesis	cells contain many chloroplasts for photosynthesis
spongy mesophyll layer		
stomata		
xylem		
phloem		

BIOLOGY ONLY

SKILLS ANALYSIS (7)

7 The graph shows the changes in the concentration of carbon dioxide in a field of long grass throughout a 24-hour period in summer.



a Explain why the levels of carbon dioxide are high at 0200 hours and low at 1200 hours.

SKILLS REASONING

b What factor will limit the rate of photosynthesis at 0400 hours and at 1400 hours?

END OF BIOLOGY ONLY

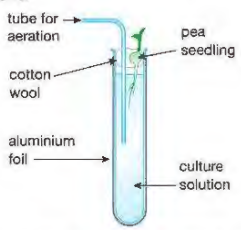
SKILLS INTERPRETATION (5)

8 The table below shows some of the substances that can be made by plants. Give one use in the plant for each. The first has been done for you.

Substance	Use
glucose	oxidised in respiration to give energy
sucrose	
starch	
cellulose	
protein	
lipid	

SKILLS REASONING (6)

9 The apparatus shown in the diagram was used to grow a pea seedling in a water culture experiment.



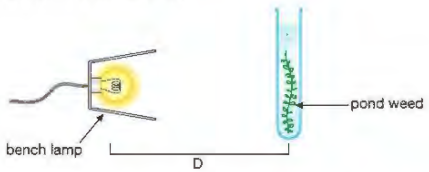
a Explain the purpose of the aeration tube and the aluminium foil around the test tube.

SKILLS REASONING (5)

b After two weeks, the roots of the pea seedling had grown less than normal, although the leaves were well developed. What mineral ion is likely to be deficient in the culture solution?

SKILLS INTERPRETATION (7)

10 A piece of Canadian pondweed was placed upside down in a test tube of water, as shown in the diagram.



Light from a bench lamp was shone onto the weed, and bubbles of gas appeared at the cut end of the stem. The distance of the lamp from the weed was changed, and the number of bubbles produced per minute was recorded. The results are shown in the table below.

Distance of lamp (D) / cm	Number of bubbles per minute
5	126
10	89
15	64
20	42
25	31
30	17
35	14
40	10

a Plot a graph of the number of bubbles per minute against the distance of the weed from the lamp.

SKILLS REASONING (6)

b Using your graph, predict the number of bubbles per minute that would be produced if the lamp was placed 17 cm from the weed.

c The student who carried out this experiment arrived at the following conclusion:

'The gas made by the weed is oxygen from photosynthesis, so the faster production of bubbles shows that the rate of photosynthesis is greater at higher light intensities.'

Write down three reasons why his conclusion could be criticised. (Hint: is counting the bubbles a reliable method of measuring the rate of photosynthesis?)

SKILLS INTERPRETATION (9)

11 Write a summary account of photosynthesis. You should include a description of the process, a summary equation, an account of how a leaf is adapted for photosynthesis and a note of how photosynthesis is important to other organisms, such as animals. You must keep your summary to less than two sides (about 500 words).